

Technical Document #918: Natural Language Processing - Comprehensive Overview

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Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Text summarization generates a concise summary of a longer text. Extractive summarization selects important sentences while abstractive summarization generates new text.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Language models predict the probability of a sequence of words.
- The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys.
- Word embeddings represent words as dense vectors in continuous space.